

Routing Concepts

Multi-vendor OSPF between FortiGate and Cisco IOS — interface configuration, adjacency, route exchange, firewall policy, and a full troubleshooting walkthrough.

Field	Detail
Author	Abdelrhman Nabil — OT / ICS Security Engineer
Platform	EVE-NG
Firewall	FortiGate-VM, FortiOS 7.6
Peer device	Cisco vIOS router
Topology	3 LAN segments, 1 shared transit network
Routing protocol	OSPF, single area 0.0.0.0

04 Lab Overview

This lab builds a routed network where a FortiGate and a Cisco router exchange routes using OSPF. It is deliberately multi-vendor: interoperability exposes the details that a single-vendor lab lets you skip.

■ Objectives

- ☐ Configure FortiGate interfaces with aliases, roles and administrative access
- ☐ Address the endpoints and verify Layer 3 connectivity to each gateway
- ☐ Configure OSPF area 0 on FortiGate and on the Cisco router
- ☐ Set LAN interfaces passive so hellos are not sent onto user segments
- ☐ Tune the transit interface — network type and DR priority
- ☐ Verify the adjacency reaches FULL and routes are exchanged both ways
- ☐ Create firewall policies to permit inter-segment traffic
- ☐ Diagnose an end-to-end failure using ICMP codes, sniffer, and debug flow

SCOPE NOTE

Dynamic routing is NOT part of the NSE 4 exam blueprint — the Routing domain covers static routing and SD-WAN only. This lab is built for practical, real-world skill and portfolio value. The concepts it does reinforce for the exam are the routing table, route selection, firewall policy behaviour, and the diagnostic workflow in Part 8.

■ Contents

Part	Section	Covers
01	Topology & addressing	Devices, interfaces, IP plan
02	FortiGate interfaces	Aliases, roles, admin access
03	Endpoint configuration	VPCS addressing, gateway verification
04	Cisco router configuration	Interfaces and OSPF process
05	FortiGate OSPF	Router ID, area, networks, passive, transit tuning
06	Firewall policies	Permitting inter-segment traffic
07	Verification	Adjacency, routing tables, end-to-end test
08	Troubleshooting	The failures encountered and how each was isolated

1 Topology & Addressing

A FortiGate and a Cisco vIOS router share a transit segment. The FortiGate owns two LAN segments, the router owns one. A Windows host on the transit network is used for management.

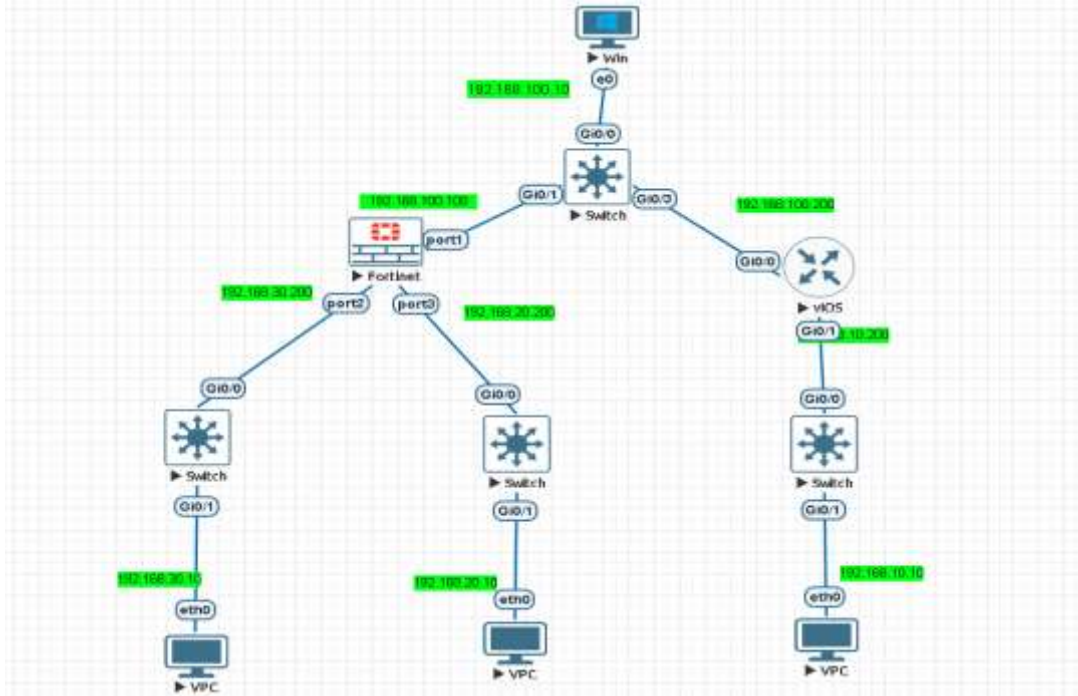


FIGURE 1 — EVE-NG LAB TOPOLOGY

■ Addressing plan

Device	Interface	Alias	Address	Segment
FortiGate	port1	Transit	192.168.100.100/24	Transit / mgmt
FortiGate	port2	30-Subnet	192.168.30.200/24	LAN-30
FortiGate	port3	20 port	192.168.20.200/24	LAN-20
Cisco vIOS	Gi0/0	—	192.168.100.200/24	Transit
Cisco vIOS	Gi0/1	—	192.168.10.200/24	LAN-10
Win-MGMT	e0	—	192.168.100.10/24	Management
VPC-30	eth0	—	192.168.30.10/24	GW .30.200
VPC-20	eth0	—	192.168.20.10/24	GW .20.200
VPC-10	eth0	—	192.168.10.10/24	GW .10.200

■ Routing design

- Protocol: OSPF, single area 0.0.0.0 (backbone)
- Router IDs: FortiGate 1.1.1.1, Cisco 2.2.2.2
- Transit type: broadcast — both devices participate in DR/BDR election
- Passive interfaces: all LAN-facing ports on both devices
- Redistribution: none — every network is advertised directly

2 FortiGate Interface Configuration

Each interface is given an address, an alias, and the minimum administrative access it needs. Aliases matter here — they appear throughout the OSPF and policy tables and make later configuration readable.

```
config system interface
  edit "port1"
    set alias "Transit"
    set ip 192.168.100.100 255.255.255.0
    set allowaccess ping https ssh
  next
  edit "port2"
    set alias "30-Subnet"
    set role lan
    set ip 192.168.30.200 255.255.255.0
    set allowaccess ping
  next
  edit "port3"
    set alias "20 port"
    set ip 192.168.20.200 255.255.255.0
    set allowaccess ping
  next
end
```

```
edit "port1"
  set vdom "root"
  set ip 192.168.100.100 255.255.255.0
  set allowaccess ping https ssh http
  set type physical
  set snmp-index 1
```

FIGURE 2 – PORT1 (TRANSIT) CONFIGURATION IN THE CLI

Name port3

Alias

Type Physical Interface

VRF ID ⓘ

Role ⓘ

☐ Dedicated Management Port

Address

Addressing mode **Manual** DHCP Auto-managed by IPAM One-Arm Sniffer

IP/Netmask

Secondary IP address ☐

Administrative Access

IPv4 ☐ HTTPS ☒ PING ☐ FMG-Access

☐ SSH ☐ SNMP ☐ FTM

☐ RADIUS Accounting ☐ Security Fabric Connection ⓘ ☐ Speed Test

FIGURE 3 – PORT3 CONFIGURED AS “20 PORT”, 192.168.20.200/24

Name	 30-Subnet (port2)		
Alias	30-Subnet		
Type	 Physical Interface		
VRF ID 	0		
Role 	LAN		

☐ Dedicated Management Port

Address

Addressing mode	☒ Manual	☐ DHCP	☐ Auto-managed by IPAM	☐ One-Arm Sniffer
IP/Netmask	192.168.30.200/255.255.255.0			
Create address object matching subnet	☐			
Secondary IP address	☐			

Administrative Access

IPv4	☐ HTTPS	☒ PING	☐ FMG-Access
	☐ SSH	☐ SNMP	☐ FTM
	☐ RADIUS Accounting	☐ Security Fabric Connection 	☐ Speed Test

FIGURE 4 – PORT2 CONFIGURED AS “30-SUBNET”, ROLE LAN, 192.168.30.200/24

3 Endpoint Configuration

Each VPCS node is given an address and a default gateway, then verified against its gateway before any routing is attempted. Testing bottom-up like this means later failures cannot be blamed on the endpoints.

```
# VPC-30
ip 192.168.30.10/24 192.168.30.200
# VPC-20
ip 192.168.20.10/24 192.168.20.200
# VPC-10
ip 192.168.10.10/24 192.168.10.200
save
```

```
VPCS> ip 192.168.30.10/24 192.168.30.200
Checking for duplicate address...
VPCS : 192.168.30.10 255.255.255.0 gateway 192.168.30.200
```

FIGURE 5 – VPC-30 ADDRESSED WITH GATEWAY 192.168.30.200

```
VPCS> ping 192.168.30.200

84 bytes from 192.168.30.200 icmp_seq=1 ttl=255 time=1.594 ms
84 bytes from 192.168.30.200 icmp_seq=2 ttl=255 time=1.471 ms
84 bytes from 192.168.30.200 icmp_seq=3 ttl=255 time=1.382 ms
84 bytes from 192.168.30.200 icmp_seq=4 ttl=255 time=1.730 ms
84 bytes from 192.168.30.200 icmp_seq=5 ttl=255 time=1.507 ms
```

FIGURE 6 – VPC-30 REACHES ITS GATEWAY; THE LOCAL SEGMENT IS HEALTHY

```
VPCS> ip 192.168.10.10/24 192.168.10.200
Checking for duplicate address...
VPCS : 192.168.10.10 255.255.255.0 gateway 192.168.10.200
```

FIGURE 7 – VPC-10 ADDRESSED WITH GATEWAY 192.168.10.200

VPCS CONFIGURATION IS NOT PERSISTENT

VPCS loses its addressing on restart unless you run `save`. A node that silently reverted to no gateway will accept pings and never reply to off-subnet sources — producing a timeout that looks like a firewall problem. Always save, and re-check with `show ip` before blaming the network.

4 Cisco Router Configuration

The Cisco side needs both interfaces addressed and an OSPF process advertising the transit and LAN networks.

```
interface GigabitEthernet0/0
 ip address 192.168.100.200 255.255.255.0
 no shutdown
!
interface GigabitEthernet0/1
 ip address 192.168.10.200 255.255.255.0
 no shutdown
!
router ospf 1
 router-id 2.2.2.2
 network 192.168.100.0 0.0.0.255 area 0
 network 192.168.10.0 0.0.0.255 area 0
 passive-interface GigabitEthernet0/1
```

```
Router#show interfaces summary
*Aug  8 20:12:40.400: %SYS-5-CONFIG I: Configured from p interface brief
Interface                IP-Address      OK? Method Status
GigabitEthernet0/0       192.168.100.200 YES manual up
GigabitEthernet0/1       192.168.10.200  YES manual up
GigabitEthernet0/2       unassigned      YES unset  administratively down
```

FIGURE 8 – SHOW IP INTERFACE BRIEF: BOTH INTERFACES UP/UP

```
Router(config)#router ospf 1
Router(config-router)#router-id 2.2.2.2
Router(config-router)#network 192.168.100.0 0.0.0.255 area 0
Router(config-router)#network 192.168.10.0 0.0.0.255 area 0
```

FIGURE 9 – OSPF PROCESS 1, ROUTER-ID 2.2.2.2, BOTH NETWORKS IN AREA 0

PASSIVE BEHAVES DIFFERENTLY TO WHAT YOU MIGHT EXPECT

On Cisco, passive-interface stops OSPF hellos on that interface but the network is STILL advertised into OSPF, provided a matching network statement exists. Passive suppresses adjacency formation, not advertisement. The same is true on FortiGate.

WILDCARD MASKS, NOT SUBNET MASKS

Cisco network statements take a wildcard mask — 0.0.0.255 for a /24, not 255.255.255.0. Getting this backwards is one of the most common reasons a network silently fails to be advertised.

5 FortiGate OSPF Configuration

Router ID, backbone area, the three connected networks, passive LAN interfaces, and explicit tuning of the transit interface.

```
config router ospf
  set router-id 1.1.1.1

  config area
    edit 0.0.0.0
      next
    end

  config network
    edit 1
      set prefix 192.168.100.0 255.255.255.0
      set area 0.0.0.0
    next
    edit 2
      set prefix 192.168.20.0 255.255.255.0
      set area 0.0.0.0
    next
    edit 3
      set prefix 192.168.30.0 255.255.255.0
      set area 0.0.0.0
    next
  end

  # LAN interfaces passive - no hellos onto user segments
  set passive-interface "port2" "port3"
end
```


Router ID

Areas

+ Create New

Edit

Delete

Area ID	Type	Authentication
0.0.0.0	Regular	None

1

Networks

+ Create New

Edit

Delete

Network	Area
192.168.100.0/24	0.0.0.0
192.168.20.0/24	0.0.0.0
192.168.30.0/24	0.0.0.0

3

FIGURE 10 – ROUTER ID 1.1.1.1, AREA 0.0.0.0, THREE NETWORKS ADVERTISED

Interfaces

+ Create New

Edit

Delete

Name	Interfaces	Cost	Apply To IP	Authentication	Passive
port2	30-Subnet (port2)	0	Any IP	None	Enabled
port3	20 port (port3)	0	Any IP	None	Enabled

2

FIGURE 11 – PORT2 AND PORT3 SET PASSIVE

■ Transit interface tuning

The transit segment is a broadcast network, so a DR/BDR election takes place. Setting the network type and priority explicitly makes the outcome deterministic rather than incidental.

```
config router ospf
  config ospf-interface
    edit "Transit"
      set interface "port1"
      set network-type broadcast
      set priority 10
    next
  end
end
```

```
Fortigate-1 # config router ospf

Fortigate-1 (ospf) # config ospf-interface

Fortigate-1 (ospf-interface) # edit "Transit"
new entry 'Transit' added

Fortigate-1 (Transit) # set interface "port1"

Fortigate-1 (Transit) # set network-type broadcast

Fortigate-1 (Transit) # set priority 10

Fortigate-1 (Transit) # next

Fortigate-1 (ospf-interface) # end

Fortigate-1 (ospf) # end
```

FIGURE 12 – TRANSIT INTERFACE: BROADCAST NETWORK TYPE, PRIORITY 10

WHY THE LANS MUST BE PASSIVE

A network statement enables OSPF on EVERY interface whose subnet matches — including LAN-20 and LAN-30, where only endpoints live. Without passive-interface, the firewall sends hellos onto user segments and will accept an adjacency from anything that answers. That is an unnecessary attack surface and, in an OT or segmented environment, an audit finding.

6 Firewall Policies

A route tells FortiGate where to send a packet. A policy decides whether it may. Both are required, and OSPF forming an adjacency proves nothing about user traffic.

```
config firewall policy
  edit 1
    set name "allow all"
    set srcintf "port2"
    set dstintf "port1"
    set srcaddr "all"
    set dstaddr "all"
    set action accept
    set schedule "always"
    set service "ALL"
    set nat disable
    set logtraffic all
  next
  edit 2
    set name "transit"
    set srcintf "port1"
    set dstintf "port2"
    set srcaddr "all"
    set dstaddr "all"
    set action accept
    set schedule "always"
    set service "ALL"
    set nat disable
    set logtraffic all
  next
end
```



FIGURE 13 – POLICIES PERMITTING TRAFFIC BETWEEN THE LAN AND THE TRANSIT INTERFACE

DISABLE NAT ON INTERNAL POLICIES

NAT is enabled by default on new policies. Between internal segments that is wrong: it rewrites 192.168.30.10 to the firewall's transit address, so the peer never sees the real source and every log entry loses attribution. Worse, it can mask a missing return route — traffic appears to work because the translated source is directly connected to the peer.

BOTH DIRECTIONS ARE REQUIRED

An ICMP echo needs a policy outbound and the echo-reply needs one inbound. A single policy produces a silent timeout that is easily mistaken for a routing failure.

Still to add: port3 (LAN-20) has no policies in the configuration above, so VPC-20 remains isolated by design. Repeat the pair using port3, or place port2 and port3 in a **zone** and write one policy pair against it — the cleaner approach as segments grow.

7 Verification

Verify in layers: adjacency first, then route exchange on both sides, then end-to-end traffic. A failure at any layer localises the fault immediately.

■ Step 1 — Transit reachability

```
execute ping 192.168.100.200
```

```
Fortigate-1 # execute ping 192.168.100.200
PING 192.168.100.200 (192.168.100.200): 56 data bytes
64 bytes from 192.168.100.200: icmp_seq=0 ttl=255 time=6.7 ms
64 bytes from 192.168.100.200: icmp_seq=1 ttl=255 time=2.1 ms
64 bytes from 192.168.100.200: icmp_seq=2 ttl=255 time=1.9 ms
64 bytes from 192.168.100.200: icmp_seq=3 ttl=255 time=2.0 ms
64 bytes from 192.168.100.200: icmp_seq=4 ttl=255 time=2.2 ms
```

FIGURE 14 – FORTIGATE REACHES THE CISCO ROUTER ACROSS THE TRANSIT SEGMENT

■ Step 2 — OSPF adjacency

```
get router info ospf neighbor
```

State FULL confirms a great deal at once: hellos in both directions, matching area, matching hello/dead timers, matching MTU and matching authentication.

■ Step 3 — Routes learned on FortiGate

```
get router info routing-table all
```

```
Fortigate-1 # get router info routing-table all
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
       O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       V - BGP VPNv4
       * - candidate default

Routing table for VRF=0
O    192.168.10.0/24 [110/2] via 192.168.100.200, port1, 00:06:11, [1/0]
C    192.168.20.0/24 is directly connected, port3
C    192.168.30.0/24 is directly connected, port2
C    192.168.100.0/24 is directly connected, port1
```

FIGURE 15 – O 192.168.10.0/24 [110/2] VIA 192.168.100.200, PORT1

The **O** code marks an OSPF route and **110** is the OSPF administrative distance. The three **C** entries are the directly connected networks.

■ Step 4 — No competing static routes

```
show router static
```

```
Fortigate-1 # show router static
config router static
end
```

FIGURE 16 — STATIC ROUTE TABLE EMPTY; NOTHING COMPETES WITH OSPF

WHY THIS CHECK MATTERS

Static routes have an administrative distance of 10 against OSPF's 110, so any leftover static route to the same prefix silently wins and OSPF never installs. Disabling redistribute static does not remove the route itself — it must be deleted.

■ Step 5 — Routes learned on the Cisco router

```
show ip route ospf
```

```
Router>show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

O       192.168.20.0/24
         [110/2] via 192.168.100.100, 00:20:54, GigabitEthernet0/0
O       192.168.30.0/24
         [110/2] via 192.168.100.100, 00:20:54, GigabitEthernet0/0
```

FIGURE 17 — THE ROUTER HAS LEARNED BOTH FORTIGATE LANS VIA 192.168.100.100

Route exchange must be confirmed in BOTH directions. A one-way exchange produces a request that arrives and a reply that never returns.

■ Step 6 — End-to-end test

```
# from VPC-30
ping 192.168.10.10
```

```
VPCS> ping 192.168.10.10

84 bytes from 192.168.10.10 icmp_seq=1 ttl=62 time=12.563 ms
84 bytes from 192.168.10.10 icmp_seq=2 ttl=62 time=11.371 ms
84 bytes from 192.168.10.10 icmp_seq=3 ttl=62 time=7.334 ms
84 bytes from 192.168.10.10 icmp_seq=4 ttl=62 time=11.802 ms
84 bytes from 192.168.10.10 icmp_seq=5 ttl=62 time=16.027 ms
```

FIGURE 18 — END-TO-END SUCCESS: REPLIES WITH TTL 62

READING THE TTL

VPCS sends with TTL 64. A reply at 62 means two routing hops — the FortiGate and the Cisco router — which is exactly the expected path. TTL is a free sanity check that traffic took the route you think it did.

8 Troubleshooting Walkthrough

The lab did not work first time. The sequence of failures below is the most valuable part of the exercise — each had a distinct signature that narrowed the fault before any configuration was changed.

■ Failure 1 — Network unreachable

```
VPCS> ping 192.168.10.10

192.168.10.10 icmp_seq=1 timeout
192.168.10.10 icmp_seq=2 timeout
192.168.10.10 icmp_seq=3 timeout
192.168.10.10 icmp_seq=4 timeout
192.168.10.10 icmp_seq=5 timeout
```

FIGURE 19 — EARLY END-TO-END TEST FAILING

The first attempts returned ICMP type 3, code 0 — network unreachable — generated by the FortiGate itself. That message only appears when the device has no route to the destination, which immediately eliminated firewall policy as a cause.

Ping symptom	Layer at fault	Where to look
ICMP type 3 code 0 (unreachable)	Routing	Routing table, OSPF adjacency, LSA exchange
Silent timeout, no reply at all	Policy or endpoint	Firewall policy, endpoint gateway
Replies with unexpected TTL	Path	Asymmetric or longer route than expected

■ Failure 2 — Silent timeout

Once OSPF installed the route, the error changed from unreachable to a plain timeout. That transition is itself the diagnosis: routing was fixed, and the fault had moved downstream to policy or the endpoints.

The cause was an endpoint — VPC-10 had no default gateway configured, so it received each echo request but could not route the reply to an off-subnet source.

■ Failure 3 — Confirming with debug flow and sniffer

```
# Trace route lookup, policy match and verdict
diagnose debug reset
diagnose debug flow filter addr 192.168.10.10
diagnose debug flow show function-name enable
diagnose debug flow trace start 40
diagnose debug enable
# ... reproduce the traffic ...
diagnose debug disable
diagnose debug flow trace stop

# Confirm packets physically enter and leave an interface
diagnose sniffer packet any "icmp and host 192.168.10.10" 4
```

Debug flow shows the route lookup, the policy match and the drop reason; the sniffer shows whether the packet ever reached the wire. Used together they distinguish an internal drop from an endpoint that is not replying.

■ Diagnostic reference

Command	Answers
get router info ospf neighbor	Is the adjacency FULL?
get router info ospf interface	Which interfaces run OSPF; which are passive; DR/BDR role
get router info ospf database	Which LSAs were received; is the peer advertising the prefix?
get router info routing-table all	What is installed in the FIB
show router static	Is a static route beating OSPF?
diagnose sniffer packet <intf> "icmp" 4	Did the packet enter or leave that interface?
diagnose debug flow	Route lookup, policy match, drop reason
diagnose sys session clear	Clears stale sessions created while routing was broken
show ip route ospf (Cisco)	Did the peer learn our networks?